

Sai Sanjay Bommisetty

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EDUCATION

University of Illinois Urbana-Champaign

B.S., Mechanical Sciences and Engineering, Hoeft T&M Minor

• **GPA:** 3.9/4.0

• **Achievements:** 1st Place (2026) & 2nd Place (2025) Engineering Industry Impact Award, 1st Place Distinguished Tech Award, Engineering Visionary Scholarship, Dean's List, Summa Cum Laude

Aug 2024 - May 2028

Champaign, IL

EXPERIENCE

International Motors

Human Factors & Mechanical Design Intern

- Developed a CV-based gaze tracking system achieving 2° accuracy via PCCR and stereo triangulation across synchronized multi-camera capture, mapping gaze onto a Unity-reconstructed 3D cab environment at 20 FPS
- Designed custom enclosures and hardware housings in Siemens NX, fabricated via 3D printing with 6-DOF adjustable mounts and cable routing to meet simulator packaging constraints
- Supported Experience Design and Clay Studio by selecting materials and planning layouts in Figma, enabling the construction of concept sleeper-cab and interior mockups that received approval for downstream prototyping
- Analyzed human factors data from 30 participants across workshops and clinic studies, synthesizing findings into thematic groupings and design concepts

May 2026 - Aug 2026

Lisle, IL

Northrop Grumman Innovation Lab

Research and Development Intern

- Designed and prototyped 3D CAD hardware housings using SolidWorks, integrating a Raspberry Pi, sensors, and bone-conduction transducers; validated electromechanical integration and sensor reliability through iterative build cycles, confirming consistent performance and reducing redesign time
- Led a 4-person team to develop a wearable hardware system that combined AR optics, custom PCB electronics, and sensor-fusion algorithms; managed end-to-end mechanical design, rapid prototyping, and physical validation testing, delivering a functional prototype that met target accuracy and usability criteria

May 2025 - Oct 2025

Baltimore, MD

LASSI – Laboratory for Advanced Space Systems at Illinois

Undergraduate Structures/Thermal Researcher

- Conducted FEA studies on the MonARCH 12U CubeSat deployer interface in Siemens NX, simulating structural loads and validating interface compliance with CubeSat standards
- Convert CAD geometry into Thermal Desktop models to evaluate temperature gradients and heat flux across satellite subsystems during orbital cycles, supporting systems engineering under SWaP and radiation constraints and delivering thermal insights that guide design improvements

Sep 2025 - Present

Champaign, IL

LEADERSHIP & EXTRACURRICULARS

American Society of Mechanical Engineers (ASME)

Carle-CIMED Product Development Team Lead

- Leading a team of 15 developing a smart mattress topper with embedded pressure sensor arrays and pneumatic actuation systems to prevent pressure ulcers affecting 2.5M+ U.S. patients; prototyping data collection systems using Arduino IDE, PCB design, and electrical engineering principles
- Designed and tested a pneumatic repositioning system projected to reduce nurse workload by 30%, earning 1st and 2nd place for the Engineering Industry Impact Award among 100+ engineering projects

Sep 2024 - Apr 2026

International Consulting Network (ICON)

Project Manager & Membership Director | PIMIC AI, Fassforward, HTS Europe, MODO

- Led consulting engagements across 4 international clients in AI, hardware, and manufacturing sectors; conducted market research, go-to-market analysis, and presented findings to stakeholders across multiple countries; leading membership recruitment and onboarding initiatives to grow and develop the consultant pipeline

Jan 2025 - Present

Open Source 4-Fingered Robotic Hand – UIUC Engineering Open House

Mechanical Team Member

- Designed and built an open source 4-fingered robotic hand for magnetic control over microbots, achieving 0.5mm positional accuracy; integrated Arduino/ESP32 microcontrollers to coordinate the full electromechanical system; earned 1st Place Distinguished Tech Award among 100+ projects

Jan 2026 - Apr 2026

TECHNICAL SKILLS

- **Software & Tools:** Python, Siemens NX, Fusion 360, Ansys Thermal Desktop, AutoCAD, SolidWorks, Blender Arduino IDE, OpenCV, ProtoPie, Figma, Unity, Microsoft Office
- **Hardware:** PCB Design, Circuit Design, Sensor Integration, Embedded Systems (ESP32/Raspberry Pi), FEA, EmotiBit
- **Interests:** Humanoid Robotics, Space Exploration, Medical Devices, Physics, Drawing, Skiing, Basketball, Hiking